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Scaling of the HC

Angular momentum conserving

$$\Phi_H \sim \left(\frac{gH\Delta_h}{\Omega^2 R_E^2} \right)^{1/2} \quad (1)$$

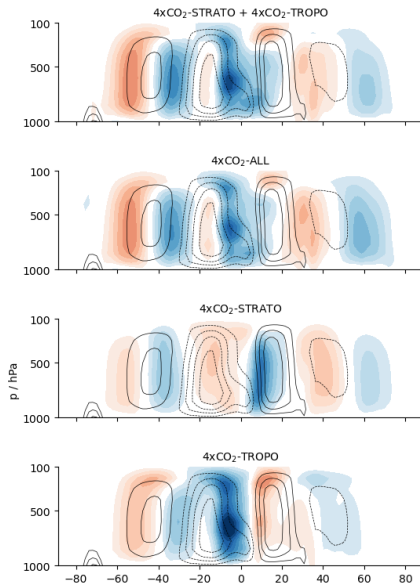
$$\psi_{\max} \sim \frac{1}{\tau} \frac{g^{3/2} (H\Delta_h)^{5/2}}{\Omega^3 R_E^2 \Delta_v} \quad (2)$$

Baroclinic instability and eddy flux

$$\theta_C \sim \left(\frac{NH}{2\Omega a} \right)^{1/2} \quad (3)$$

$$\psi(1 - Ro) \approx \frac{S}{f} \quad (4)$$

ECHAM mean meridional circulation



Definitions D'Agostino et al. 2017

- meridional near-surface(averaged between the 700 and 800 hPa levels) potential temperature contrast
$$\Delta_h\theta(p, \varphi) = \bar{\theta}(800 - 700 \text{ hPa}, 0 - 10^\circ) - \bar{\theta}(800 - 700 \text{ hPa}, 40 - 60^\circ)$$
- subtropical near-surface static stability
$$\Delta_v\theta(p, \varphi) = \bar{\theta}(700 \text{ hPa}, 20 - 40^\circ) - \bar{\theta}(800 \text{ hPa}, 20 - 40^\circ)$$
- tropical tropopause pressure height $p_t(\varphi) = \bar{p}(15^\circ\text{S} - 15^\circ\text{N})$
- extratropical tropopause height $p_{et}(\varphi) = \bar{p}(35^\circ - 55^\circ)$ (Lu et al. 2007)

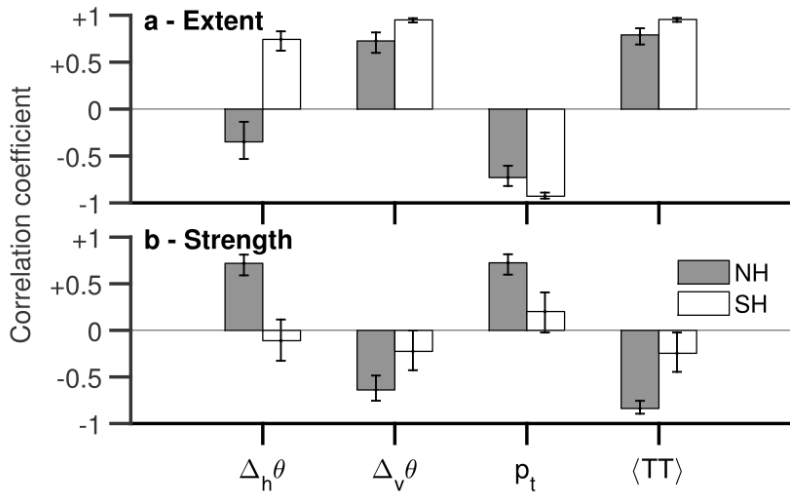
Trends in NH from ECHAM - strength

	PI	4x Tropo	4x Strato	Uniform
$\Delta_h \theta$	16.5 K	15.9 K (-)	16.0 K (-)	15.4 K (-)
$\Delta_v \theta$	1.61 K	1.94 K (-)	1.75 K (-)	2.09 K (-)
p_t	99.4 hPa	83.6 hPa (+)	105.0 hPa (-)	87.4 hPa (+)
p_{et}	207.8 hPa	183.3 hPa (+)	212.4 hPa (-)	184.3 hPa (+)
HC strength		(+)	(-)	(-)

Trends in NH from ECHAM - width

	PI	4x Tropo	4x Strato	Uniform
$\Delta_h \theta$	16.5 K	15.9 K(-)	16.0 K (-)	15.4 K (-)
p_t	99.4 hPa	83.6 hPa (+)	105.0 hPa (-)	87.4 hPa (+)
p_{et}	207.8 hPa	183.3 hPa (+)	212.4 hPa (-)	184.3 hPa (+)
HC width		(+)	($\pm$)	(+)

Winter correlation D'Agostino et al. 2017



Equinox as annual mean

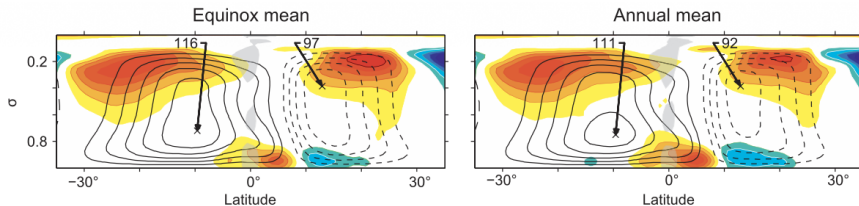


Figure: Levine & Schneider 2010

Eddy fluxes

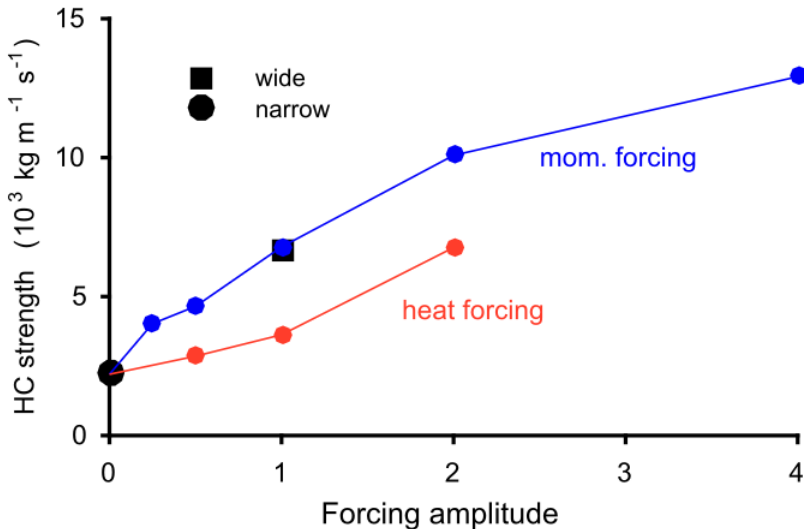


Figure: Singh et al. 2017

Atmospheric Study Project Colloquium - 02.02.2026

- Investigating the strengthening of the Hadley cell under tropospheric-only CO₂ forcing
- The strengthening of the Hadley circulation (cell) driven by CO₂ forcing confined to the troposphere
- The impact of tropospheric versus stratospheric CO₂ forcing on the Hadley cell

- Motivate and explain your topic!
- Present the state-of-art concerning your topic: What is already known? Which results are already published? Are there publications of special relevance for you, e.g. by serving partly as a blue print?
- Which research gap do you want to fill? What are the innovations? What are your specific research questions and/or hypothesis?
- Which tools, methods, data etc. do you plan to use?
- Outline your workplan!
- (If possible, you can also present first preliminary results.)